CONCEPT BUSINESS PROPOSAL

The Vertical Barn



User-supplied architectural inspiration. The tower, trees and rail are illustrative; the model below is a separate six-level pilot, not a measured reconstruction.

Footprint	Gross floor area	Crop canopy
1,200 m²	7,200 m² / 77,500 ft²	7,800 m² / 84,000 ft²

Proposition. Operate a barn-shaped controlled-environment farm supplying premium herbs and leafy greens to regional foodservice and specialist retailers. Stack accessible production floors above a ground-floor packing and services hub.

Investment view. Author-assumed base case: £3.28m annual revenue, £0.56m EBITDA and £18m initial funding. That implies about 32 years of simple EBITDA payback, before replacement capital and financing. Fund a small validation phase first; do not commit to the tower on these economics.

UK/England working location; all financial inputs are illustrative in 2026 GBP, excluding VAT. No site, customers, quotations or investment commitments have been verified.

What the extra space means

Square metres (m²) are used throughout. Floor area, usable area and planted canopy are different measures.

Area measure	Calculation	Result
Building footprint	40 m × 30 m	1,200 m²
Gross internal floor area	6 floors × 1,200 m²	7,200 m²
Increase over one floor	7,200 - 1,200	+6,000 m² / +500%
Usable internal area	75% × 7,200	5,400 m²
Production floor space	5 floors × 780 m²	3,900 m²
Other usable space	Ground services + floor support	1,500 m²
Walls, cores and circulation	25% × 7,200	1,800 m²
Planted canopy	3,900 m² × 2 growing tiers	7,800 m²
Site area allowance	Building + yard/access/drainage	4,000 m²

Like-for-like comparison

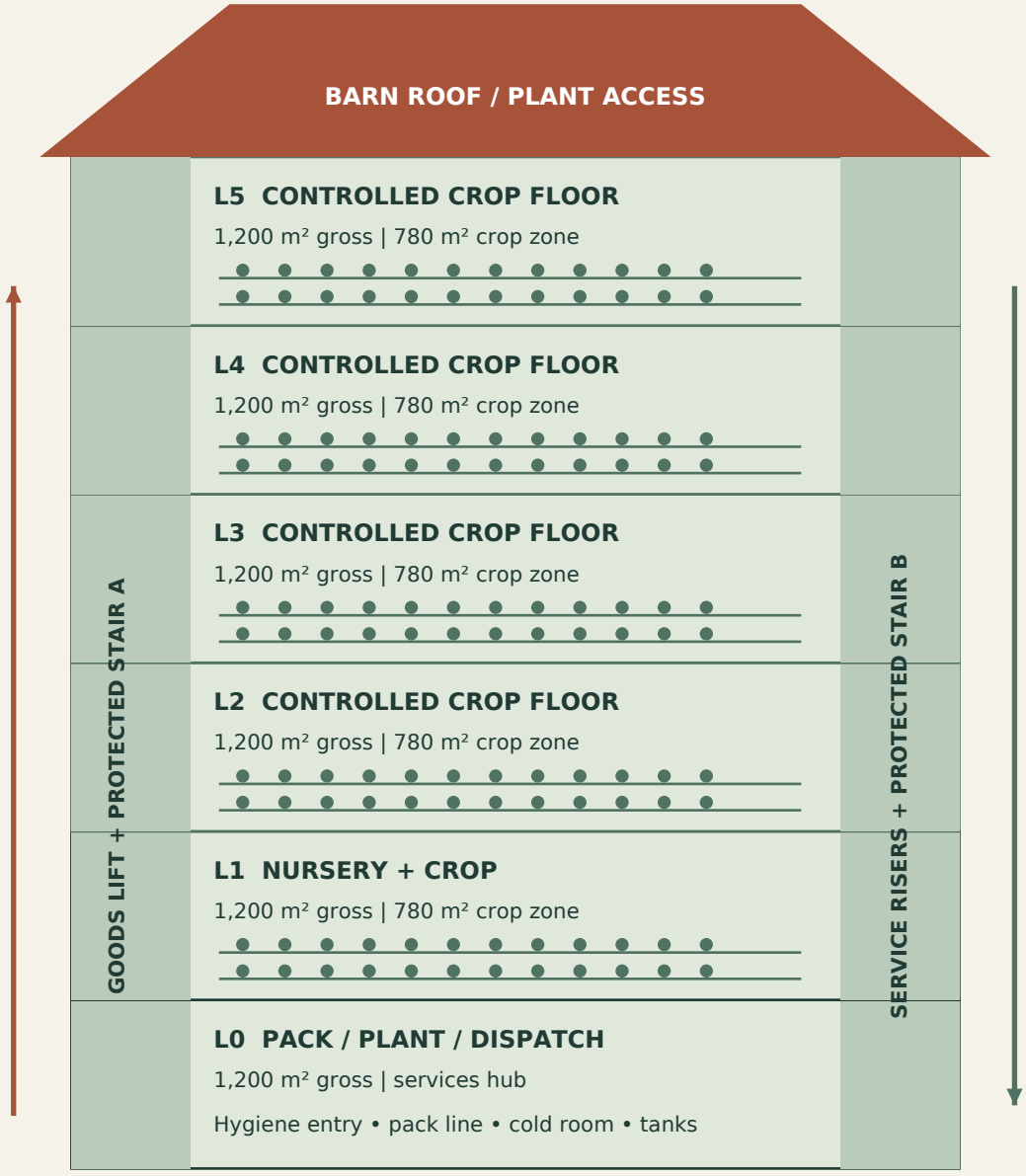
A single floor with the same 65% crop allocation provides 780 m² of canopy at one tier, or 1,560 m² at two tiers. The proposal offers 10 times the one-tier canopy, or 5 times the two-tier canopy, on the same building footprint. It does not create 10 times the land or guarantee 10 times the output.

Critical alternative: a single-storey high-bay shed with ten rack tiers on 780 m² could also provide 7,800 m² of canopy. Compare equal canopy, yield and annual output before choosing the multi-floor tower. Accessible floors improve compartmentation and maintenance access but add slabs, cores, lifts and structural cost.

Whole-site canopy density is 7,800 / 4,000 = 1.95 m² canopy per m² of site. The building footprint occupies 30% of the assumed site. Roof attic, landscaping, balconies and rail infrastructure earn no area or income credit.

How the multi-level barn works

Schematic section - not to scale. Five crop floors above a services base; independent floor systems limit shared failures.



UP: clean trays, inputs and trained staff

DOWN: sealed harvest totes to packing and cold dispatch

Water: ground treatment and make-up water feed isolated floor loops. Each loop doses nutrients and recirculates locally; monitored purge goes to a permitted treatment/disposal route. Condensate is reused only after water-quality validation.

Air and safety: each growing zone has climate control and filtered air. Two separate protected escape routes are shown conceptually; goods lifts are not escape routes. Separate hygiene schedules control clean and dirty movements.

From seed to dispatch

Commercial scope: small, high-value crops. No livestock, tree orchards or staple grains in the financial model.

1 / Receive and propagate

Record seed lots, inspect inputs and germinate in a segregated nursery on L1. Nursery occupancy is already included in the 780 m² floor crop allowance; annual yield is measured across the whole canopy, including nursery and downtime.

2 / Move and grow

Move clean trays in scheduled lift slots to L1-L5. Two-tier racks use crop-specific LED recipes, irrigation, temperature and humidity controls. Staff access each floor directly; sensors report leaks, power draw, climate and crop condition.

3 / Harvest and maintain separation

Harvest by batch into closed food-safe totes. Descend to the ground packing hub in dedicated time windows. Waste, used trays and maintenance equipment follow a segregated return schedule and cleaning procedure.

4 / Pack, chill and deliver

Inspect quality, pack to buyer specification, record traceability and chill appropriately for the crop. Dispatch daily to a short regional delivery radius. Buyer contracts define grade, rejection limits, volumes and payment terms.

5 / Recover and verify resources

Track electricity and water per saleable kilogram. Reuse recirculating water within isolated zones; assess recovered heat and condensate. Do not claim zero water use, zero pesticides or carbon neutrality without measurement.

Engineering decisions before tender

Allow roughly 4.5 m floor-to-floor as an early assumption: six levels imply about 27 m to eaves plus the barn roof. Engineers must determine slab/rack loads including saturated systems, vibration, wind, fire strategy, corrosion protection and drainage. Keep heavy water storage at ground level. Mature facade trees in the reference are replaced by modest maintainable planting.

Grid demand: base annual use is 4.095 GWh, or 467 kW average; peak connection demand requires an hourly load study. Roof solar cannot be assumed to supply the full farm. Power, pump and cooling resilience must be specified around crop survival times.

Customers before construction

Operate the first barn directly. Replication or licensing begins only after a full production year demonstrates repeatable performance.

Target buyer hypothesis: regional restaurant distributors, premium grocers and food manufacturers needing consistent herbs and specialist leaves. Validate buyers within the chosen delivery radius. No market share, demand volume or premium price is assumed to be independently established.

Contract structure: weekly standing orders, minimum volumes, agreed grading and packaging, indexed energy clauses where buyers accept them, and several customers to limit concentration. The model uses net producer receipts, after discounts, rather than supermarket shelf prices.

Illustrative initial funding requirement

Category	£ million
Site acquisition / preparation allowance	0.80
Structure, floors, facade and roof	6.00
LED racks, irrigation and crop equipment	3.00
Climate systems, water and electrical installation	2.20
Lifts, fire systems, packing and cold room	1.40
Design, surveys and permissions	0.80
Contingency	2.30
Pre-opening, ramp losses and working capital	1.50
Total initial funding	18.00

Order-of-magnitude author allowances, not contractor estimates. Land, utilities and abnormal ground conditions can materially change the total. Contingency is approximately 16% of the £14.2m pre-contingency development subtotal. No grants, rail link, visitor income, carbon credits or facade crops are credited.

Preferred location and differentiation

Seek an accessible industrial-edge or established farm-business site near buyers, suitable grid capacity and an economic heat/cooling source. Differentiate on reliable specification, freshness and traceability. A scenic hillside is an architectural reference, not the default location for a costly production asset.

Does the added area pay?

Stabilised annual operation, illustrative 2026 GBP. These are underwriting assumptions to test, not forecasts or market quotations.

Base input	Assumption
Saleable output	7,800 m² × 35 kg/m²/year = 273,000 kg
Net realised selling price	£12.00/kg
Electricity intensity and tariff	15 kWh/kg × £0.20/kWh = £3.00/kg
Other variable costs	Inputs £1.30 + packing/delivery £1.50 + other £0.40/kg
Fixed operating costs	Payroll £720k + other fixed overhead £300k
Payroll basis	18 FTE × £40k fully loaded (assumption)

Annual base case	£ million
Revenue	3.2760
Variable costs (£6.20/kg)	(1.6926)
Contribution (£5.80/kg)	1.5834
Fixed operating costs	(1.0200)
EBITDA / margin	0.5634 / 17.2%
Illustrative depreciation (£16.5m / 20 years)	(0.8250)
Illustrative EBIT	(0.2616)

Cash operating break-even: £1.02m / £5.80 = 175,862 kg/year, or 64.4% of base saleable output. Average cash operating cost is £9.94/kg; incremental cost is approximated at £6.20/kg only within existing capacity. Neither includes capital recovery or financing.

Scenario	Downside	Base	Upside
Yield kg/m²/yr; price/kg	30; £8	35; £12	45; £14
Energy kWh/kg; £/kWh	20; £0.25	15; £0.20	12; £0.15
Annual EBITDA	-£1.067m	£0.563m	£2.139m
Simple £18m / EBITDA payback	None	32.0 years	8.4 years

All scenarios use the same canopy, £3.20/kg non-energy variable cost and £1.02m fixed cost; higher output may require extra staffing, not modelled. Each 1p/kWh adds £40,950 annual energy cost at base volume; each £1/kg price change moves EBITDA by £273,000. Payback is not an IRR: it excludes tax, debt, replacements and ramp-up.

Prove the farm, then build

Recommendation: approve a bounded feasibility programme, not unconditional tower construction.

Phase	Indicative timing	Evidence to proceed
Feasibility	Months 0-3	Site shortlist; buyer interviews; concept engineering; compare tower, high-bay racks and greenhouse.
Demonstration	Months 3-12	Existing leased premises; 200-400 m² canopy; crop trials, labour logs, measured kWh/kg and paid repeat orders.
Investment decision	Months 12-15	Independent cost plan, power offer, permissions strategy, buyer commitments and credible downside cash cover.
Design / build / commission	Months 15-36+	Stage construction, test systems and ramp crop rooms sequentially; timing depends on consent and grid connection.

Feasibility envelope: initially cap desktop/site/buyer work at an assumed £75k. Obtain quotations before separately funding the demonstration; neither stage constitutes an authorisation to spend. Reconcile these early costs with the £18m budget to avoid double counting.

Investment hurdles

Demonstrate repeat sales at the modelled net price; validate full-facility energy and labour, not lighting alone; quantify crop losses and buyer rejections. Seek an unlevered stabilised EBITDA yield above 10% as a screening hurdle (a sponsor choice, not a market norm). The base model achieves just 3.1%; the upside reaches 11.9% before replacement capital.

Key risks and responses

Planning and structure: the assumed height exceeds the usual 12 m Class A agricultural permitted-development limit away from aerodromes in England [2]. Obtain site-specific planning and building-control advice; no consent is assumed.
Crop and energy: use isolated zones, tested alarms and supply resilience; contract energy only after modelling loads.
Market: trial orders and multiple buyers before scale; do not price output as retail packs.
Cost and operations: independent cost validation, maintenance access, replacement reserves and a volume-linked staffing plan.

Prepare food-safety and traceability procedures and confirm the applicable registration route; FSA guidance requires food-business registration at least 28 days before trading in England [3]. Obtain early authority advice for this farm and packing operation.

What is known, what must be tested

Sources checked 11 November 2026. References support context and process requirements; they do not validate the model inputs.

[1] UK Parliament - energy exposure

House of Lords, Sowing the seeds: A blooming English horticultural sector, chapter 2 (2023), paragraphs 36-44. Describes energy intensity and exposure to price fluctuations. Historical policy details are not treated as current support entitlements.
publications.parliament.uk / horticultural sector / chapter 2

[2] England agricultural development rules

General Permitted Development Order 2015, Schedule 2, Part 6, Class A, A.1(g), version accessed for 2026. Sets the 12 m height limitation outside the specified aerodrome proximity. GOV.UK advises checking with the local planning authority before relying on farm permitted-development rights.
legislation.gov.uk / SI 2015/596 / Schedule 2 / Part 6
gov.uk / planning-permissions-for-farms / permitted-development

[3] Food Standards Agency - registration

Food business registration, GOV.UK, updated 25 June 2026. Registration timing and local authority route; scope must be confirmed for the chosen operation.
gov.uk / guidance / food-business-registration

Assumptions register

All dimensions, floor efficiencies, tier counts, yields, selling prices, tariffs, capital costs, staffing, operating costs and financial hurdles are author-created scenarios. No supplier quotes or crop-specific test results have been obtained. Annual yield means saleable kilograms per total canopy square metre, after downtime and rejects. No additional utilisation factor is applied.

The 15 kWh/kg base energy assumption is a test target, not an industry benchmark; weaker performance is explicitly stressed. Revenue depends on the weighted crop mix securing £12/kg net receipts. Racks and LEDs do not last forever: a detailed model must add asset-specific useful lives and replacement cash flows before financing.

Concept preserved; construction rationalised

Retain the red barn roof, vertical rhythm and planted edges of the supplied images. Use a regular engineered structure and maintainable facade planting. Mature trees on stacked ledges, mountain earthworks and passenger rail are excluded. The diagram is a functional concept, not a construction, fire, water or agronomic design.

Next deliverable: a site-specific feasibility study with a buyer-backed crop plan, three comparable building options, independently priced capital budget, hourly energy simulation and a monthly cash-flow model.